

Evolutionary Collective Robotics

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Kilobots @ ISIR, Sorbonne Université

UE Apprentissage et Robotique (AROB)
M1 informatique, parcours Androïde

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Outline

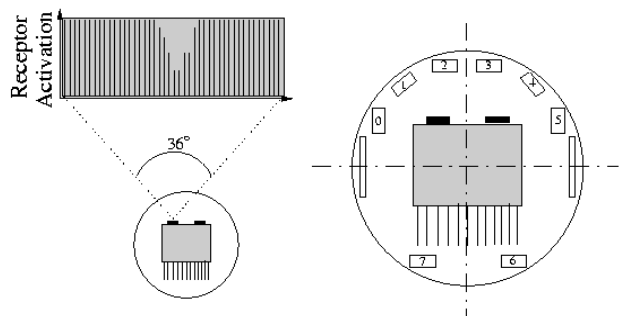
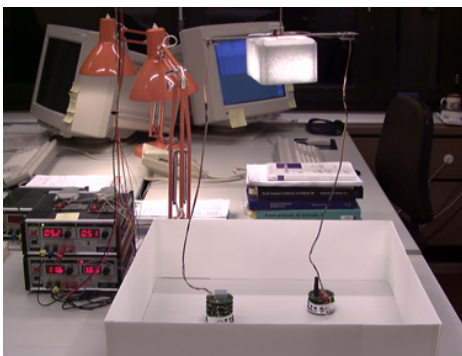
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- Evolutionary Collective Robotics
 - ▶ Competitive setups
 - ▶ Cooperative setups
 - ▶ The “clonal” approach
 - ▶ Homogeneous vs. heterogeneous population
 - ▶ A model for evolutionary biology

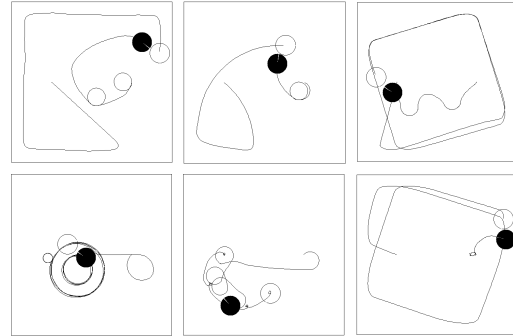
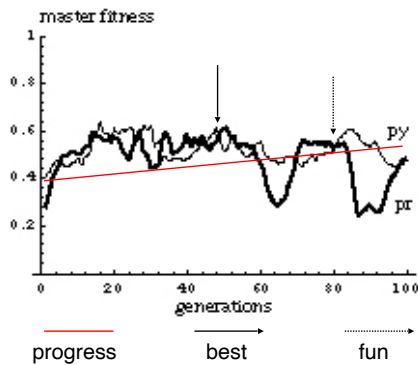
Competitive setups

Case study: predator-prey

4



Predator : camera + prox. sensors
Prey : proximity sensors + faster



Definitions



- Adaptive landscape
 - fitness changes in response to changes in the environment
- Co-evolutionary arms race
 - cyclical interaction between competing strategies
- Red Queen effect
 - continuous adaptation just for the sake of maintaining one's relative fitness (i.e. not for incremental advantage)

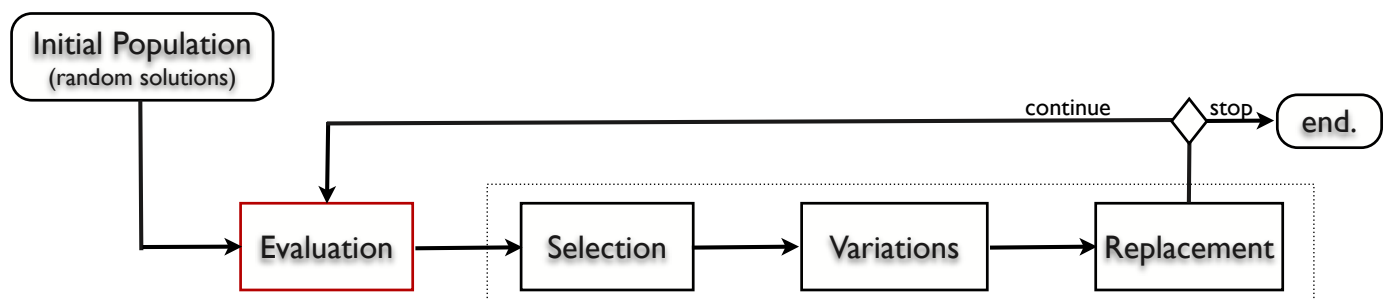
Cooperative setups (1)

The “clonal” approach

Evolution Robotics

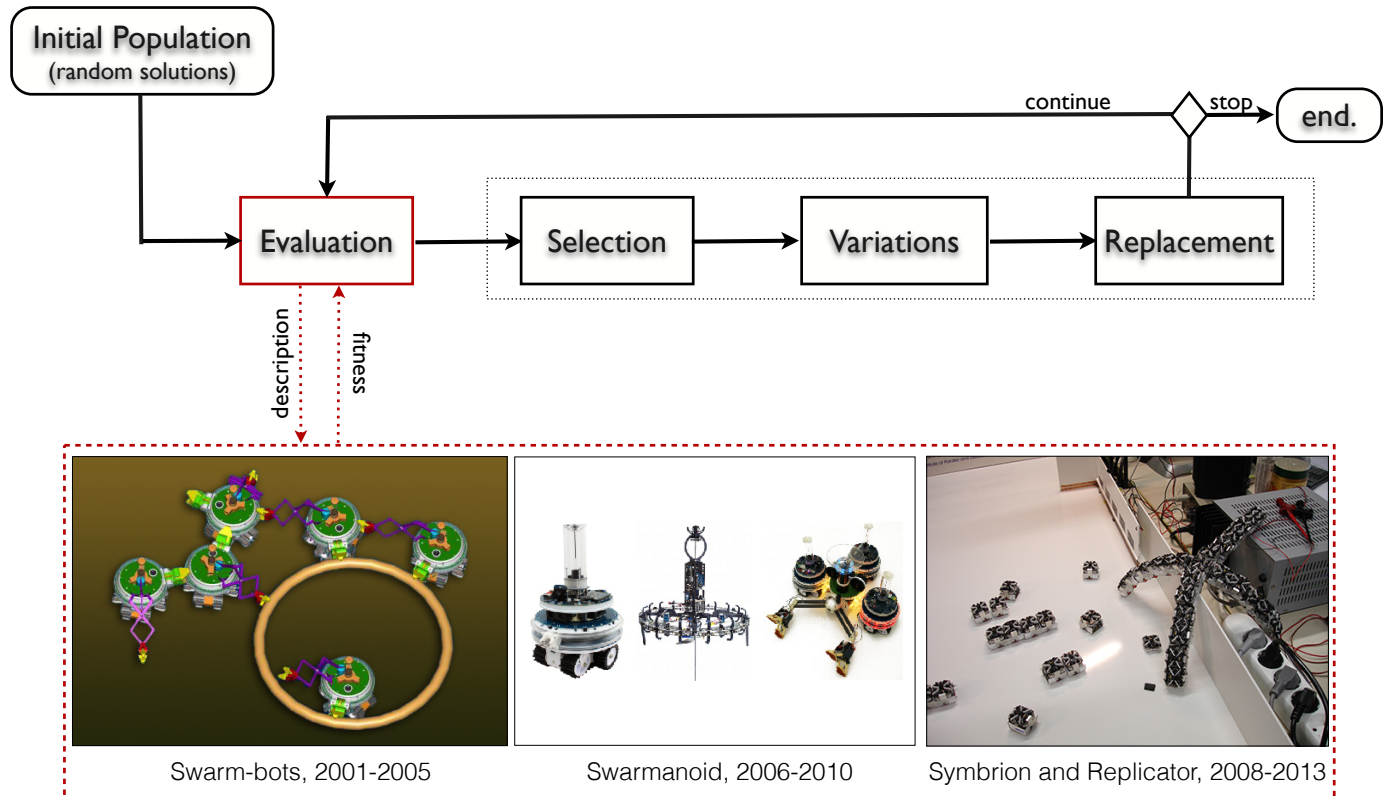
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A gentle reminder from last courses



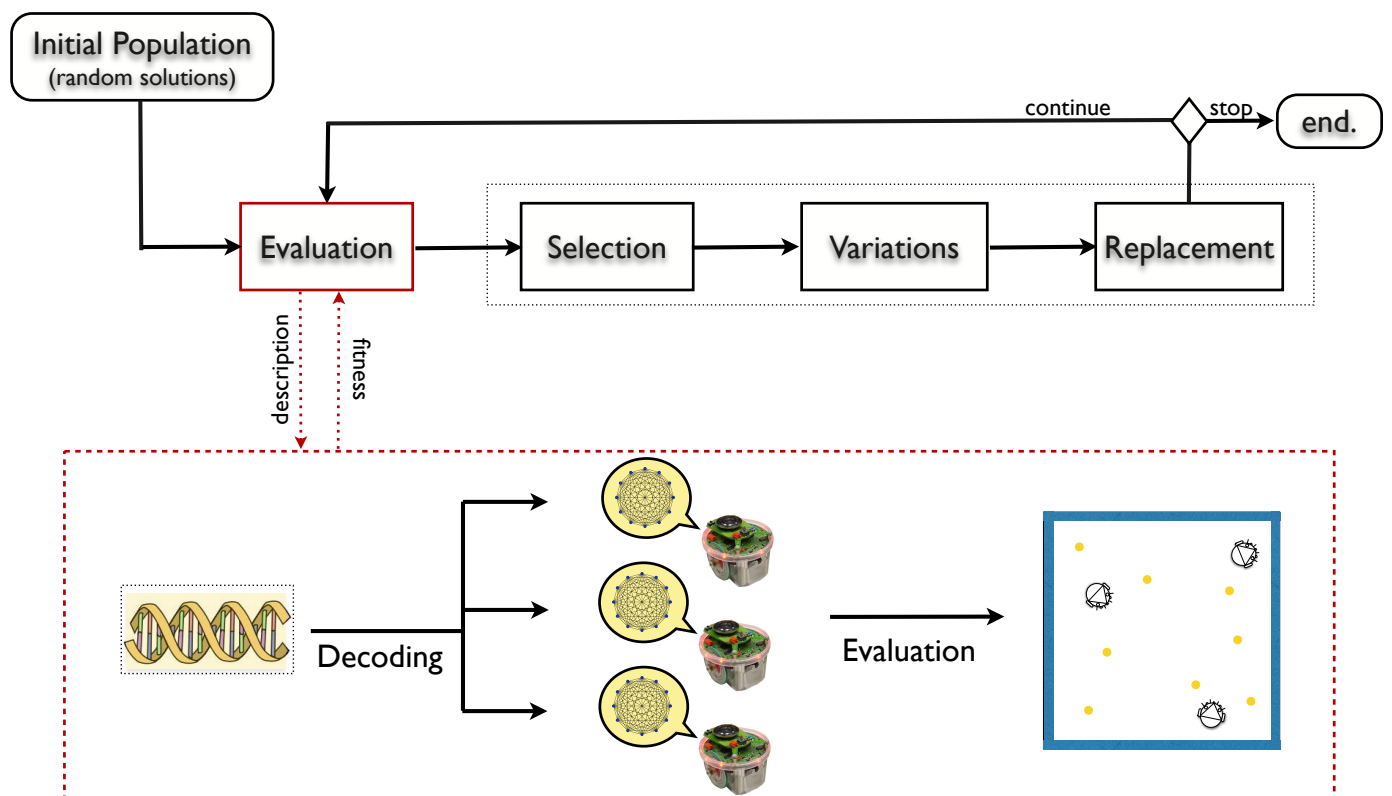
- What?
 - ▶ Off-line design method : classic “**evolutionary robotics**” method
 - ▶ Optimize in centralized fashion, **then** used in a distributed fashion
- Expected result
 - ▶ Parameters of a policy used by the robot to solve a task

for collective and swarm robotics



Homogeneous population

“clonal” approach

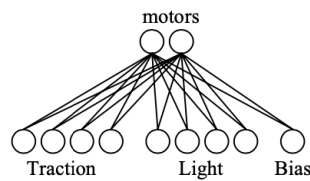


EU-funded project 2001-2005

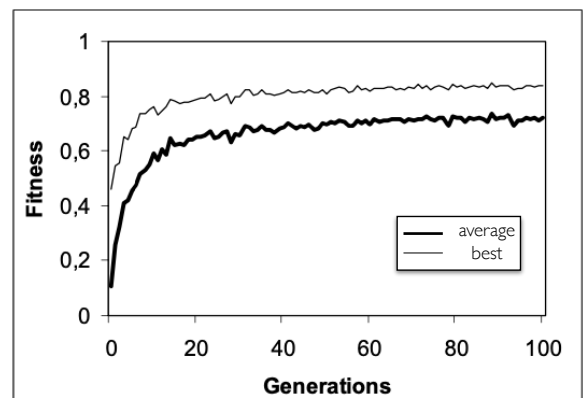
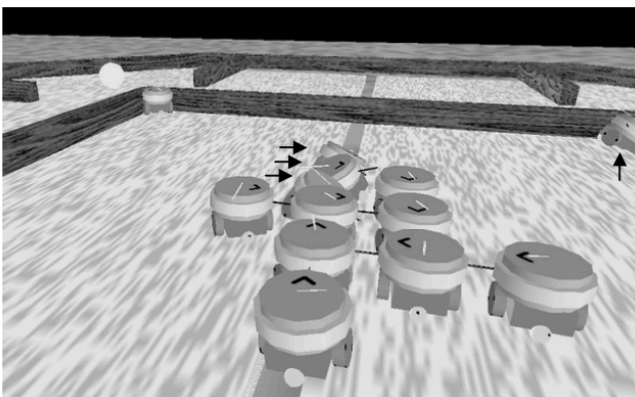


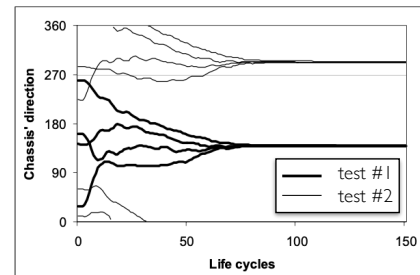
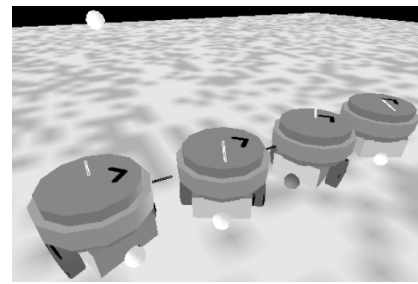
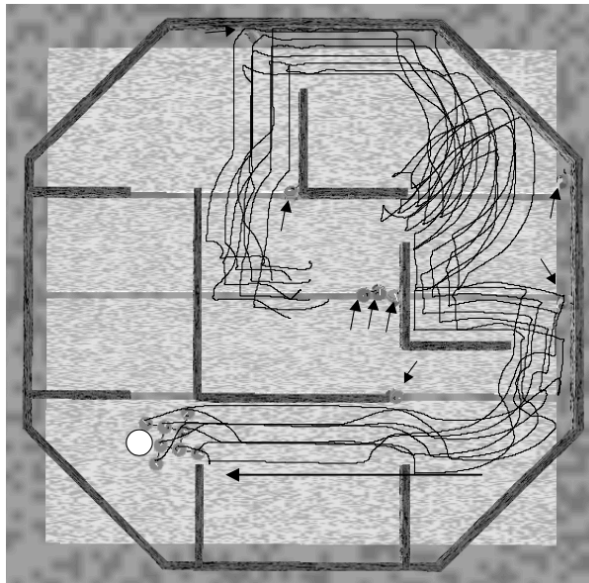
coordinated locomotion

EU-funded project 2001-2005



x #robots





[Baldassarre et al. 2006]

Task specialisation with homogeneous team

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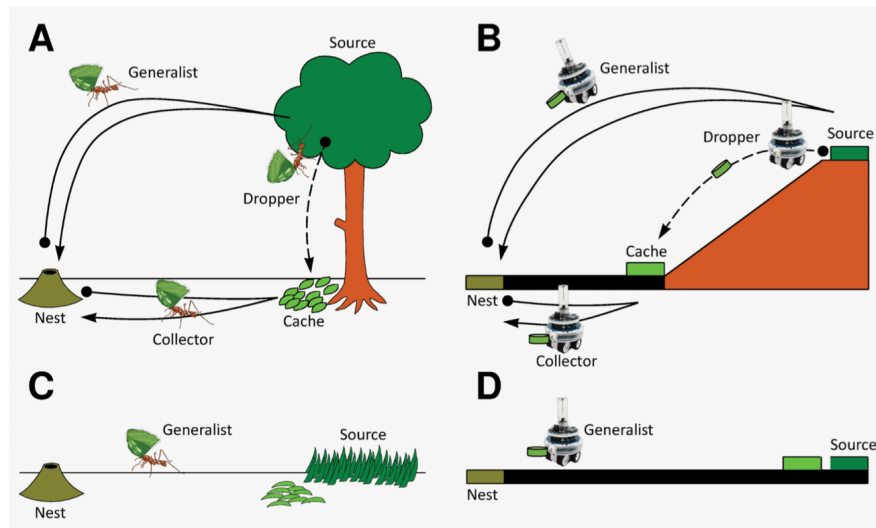
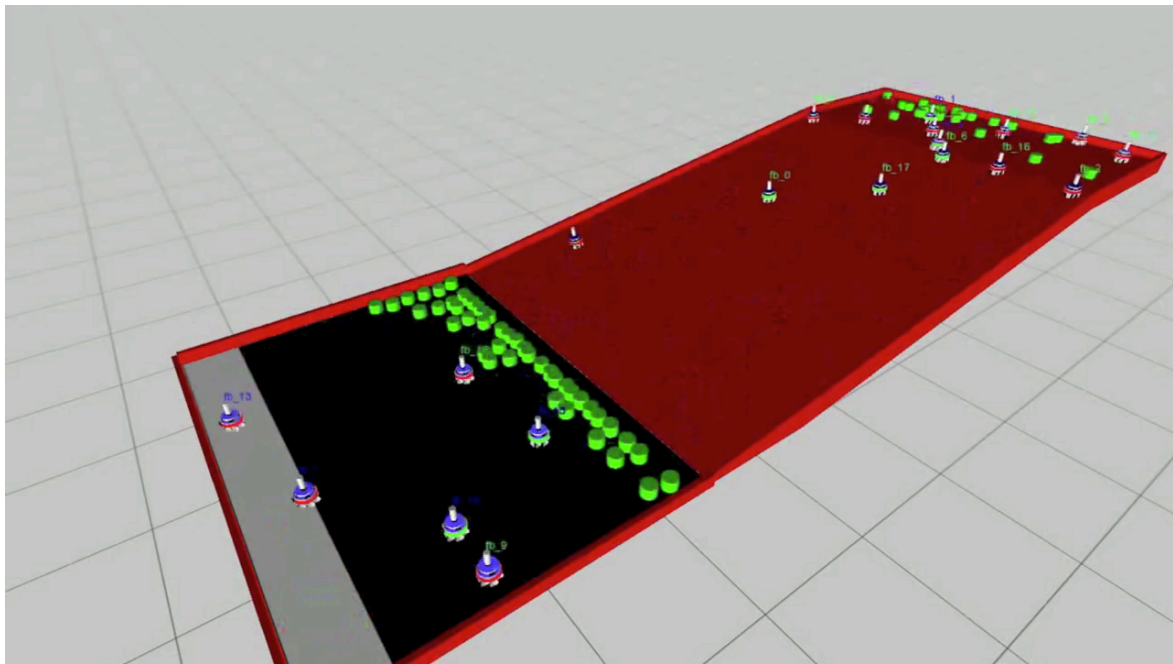


Fig 1. Task partitioning in insects and robots. (a) Task partitioned retrieval of leaf fragments, as found in most *Atta* leafcutter ants that harvest leaves from trees [7,43]. Dropper ants cut leaves which then accumulate in a cache, after which the leaves are retrieved by collectors and brought back to the nest, where they serve as a substrate for a fungus which is farmed as food. Ants also occasionally use a generalist strategy whereby both tasks are performed by the same individuals. (b) Analogous robotics setup, whereby items have to be transported across a slope using the coordinated action of droppers, collectors and possibly generalists. (c) Grass cutting leafcutter ants cutting leaf fragments in a flat environment without task partitioning, using a generalist foraging strategy [49]. (d) Analogous robotics setup, with robots being required to collect items in a flat arena.



[Ferrante et al., 2015]

<https://www.youtube.com/watch?v=8mIHxCNzjg>

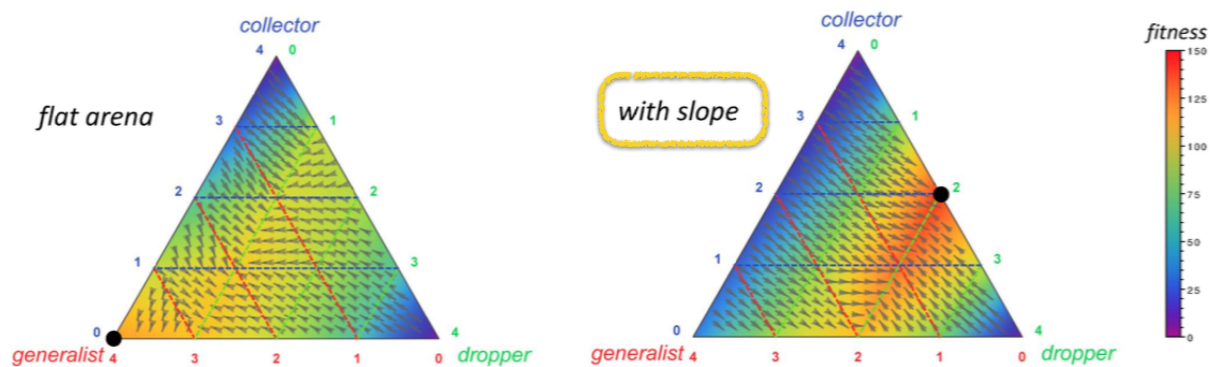
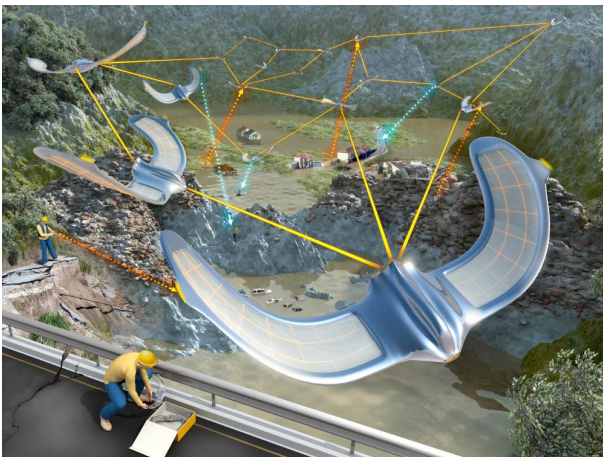


Fig 3. Optimal group composition in 4 robot teams using pre-adapted dropper, collector or generalist foraging strategies (cf. hand-coded rules shown in S1 Table). Ternary plots show group performance (total number of items retrieved to the nest over a period of 5,000 simulated seconds averaged over 10 simulation runs, color coded) as a function of the number of collectors (blue), droppers (green) and generalist foragers (red) in the 4 robot teams (black dot = optimum). In a flat environment (a), teams of generalist foragers achieve optimal performance (cf. S2 Video), whereas in a sloped arena (b), a mix of 2 droppers and 2 collectors is most optimal (cf. S1 Video). Both of these optima are global attractors in their respective fitness landscapes (cf. vectors which represent the phase portrait).

[Ferrante et al., 2015]

Take-home message

Homogeneous team can be easily optimised
Homogeneous team are capable of specialisation



Evolved swarming without positioning information: an application in aerial communication relay

Sabine Hauert · Jean-Christophe Zufferey · Dario Floreano

[Autonomous Robots](#)

January 2009, Volume 26, [Issue 1](#), pp 21–32



Evolution of Collective Behaviors for a Real Swarm of Aquatic Surface Robots

Miguel Duarte , Vasco Costa, Jorge Gomes, Tiago Rodrigues, Fernando Silva, Sancho Moura Oliveira, Anders Lyhne Christensen

Published: March 21, 2016 • <https://doi.org/10.1371/journal.pone.0151834>



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QUIZZ

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Q: En coévolution, la course aux armements provoque systématiquement un effet “reine rouge”?

1. oui
2. non

Q: l'approche clonale c'est...

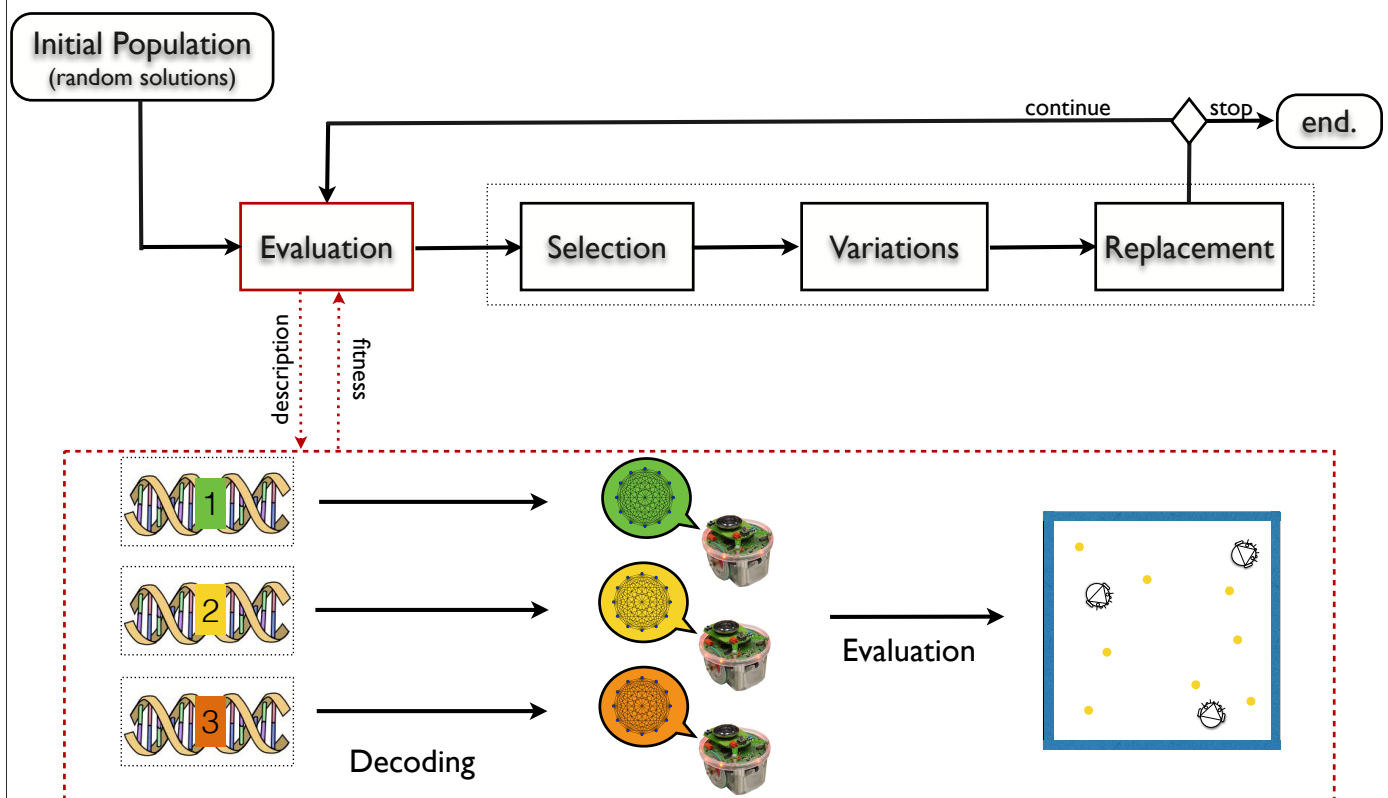
1. toutes les solutions candidates sont identiques
2. tous les robots d'un groupe sont identiques
3. les deux

Q: dans l'approche clonale, est-il possible d'observer des robots se spécialisant sur une tâche?

1. oui
2. non

Cooperative setups (2)

homogeneous vs. heterogeneous populations



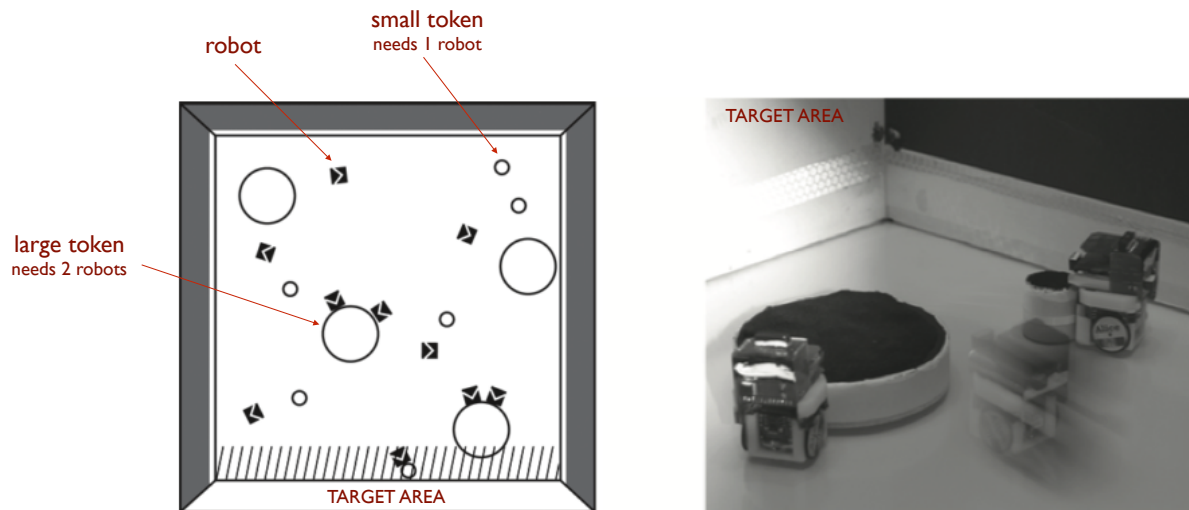


Fig. 3. Left: Experimental setup for task 3, the altruistic cooperative foraging task. Ten micro-robots (black squares with arrows) searched for small and large tokens and transported them to the target area (hatched area at bottom) under the white wall (the other three walls were black). An identical setup was used in the other two tasks, except that the arena contained either only small tokens in task 1, or only large tokens in task 2. Right: Three micro-robots in task 3, which is the altruistic cooperative foraging task. The robot in the background could transport the small token by itself. The robot at the left could not transport the large token by itself and needed to wait for the arrival of a second robot (blurred in the picture due to its rapid movement towards the large token).

Genetic Team Composition and Level of Selection in the Evolution of Cooperation

IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION, VOL. 13, NO. 3, JUNE 2009

Markus Waibel, Laurent Keller, and Dario Floreano

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Waibel et al., 2009

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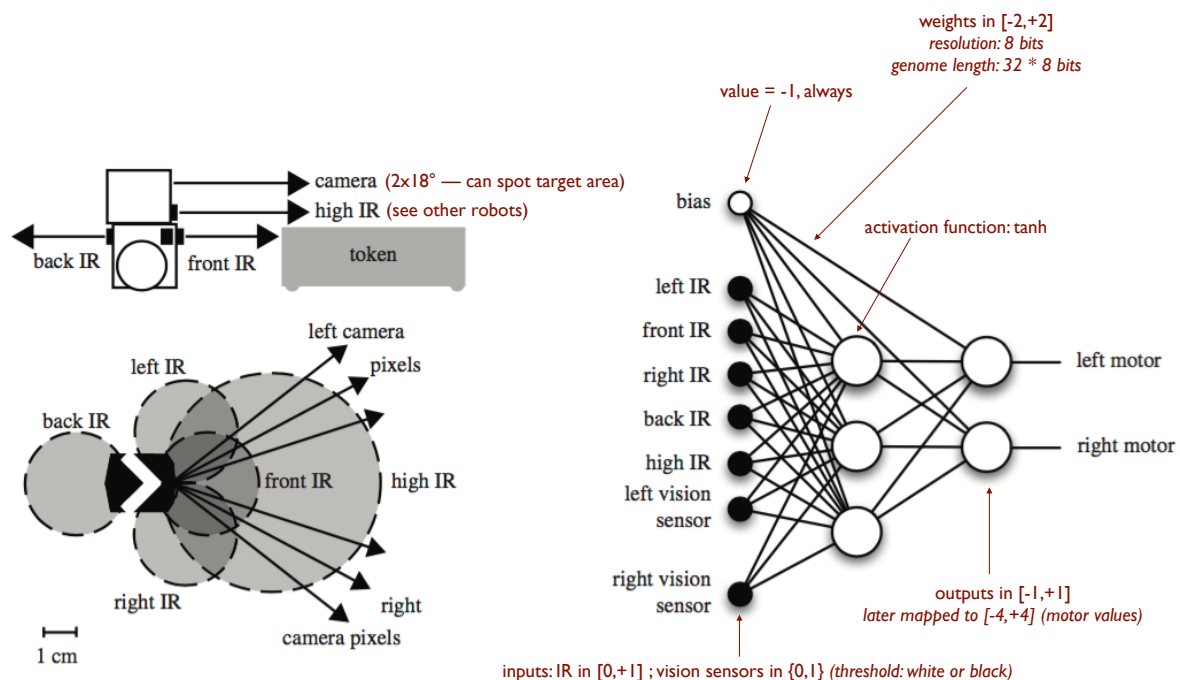
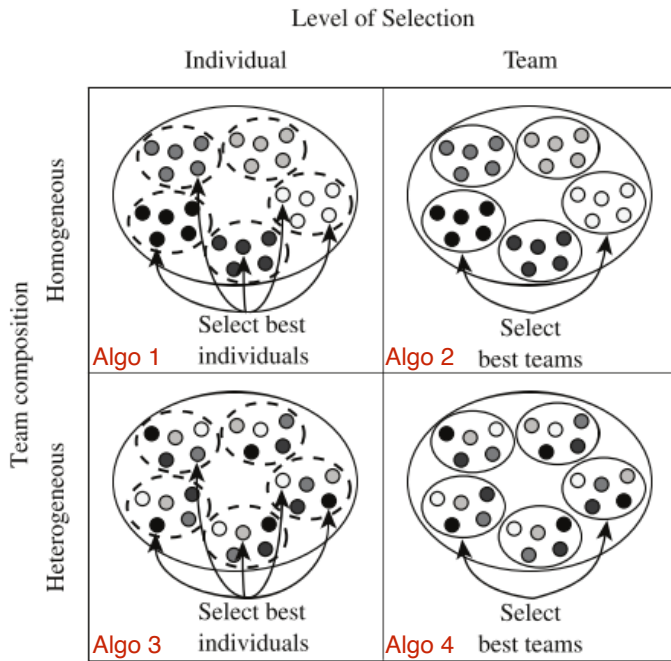


Fig. 4. Left: Side and top-view schematics of a simulated micro-robot. The robot was equipped with four infrared (IR) distance sensors (three at the front, one at the back) to detect tokens, and a camera to identify the target area. A fifth infrared distance sensor (high IR) was mounted higher on the robot and thus overlooked tokens. This allowed robots to distinguish tokens from walls and other robots. Right: The neural network architecture, which is a feed-forward neural network with a single layer of three hidden neurons. Inputs were given by the activation values of five infrared (IR) sensors and two vision sensors with activation values computed from left and right camera pixels (see text).



Algorithm 1

- a team contains clones
- assign fitness to individuals
- selection among all individuals

Algorithm 2

- a team contains clones
- assign fitness to teams
- selection among teams

Algorithm 3

- a team contains unique individuals
- assign fitness to individuals
- selection among all individuals

Algorithm 4

- a team contains unique individuals
- assign fitness to individuals
- selection among teams

$M=100$ teams, $N=10$ individuals

To generate a new individual

- roulette wheel selects 2 parents
- variation: one-point Xover + bit flips

Fitness is averaged over 10 (independent) evaluations

[Waibel et al., 2009]

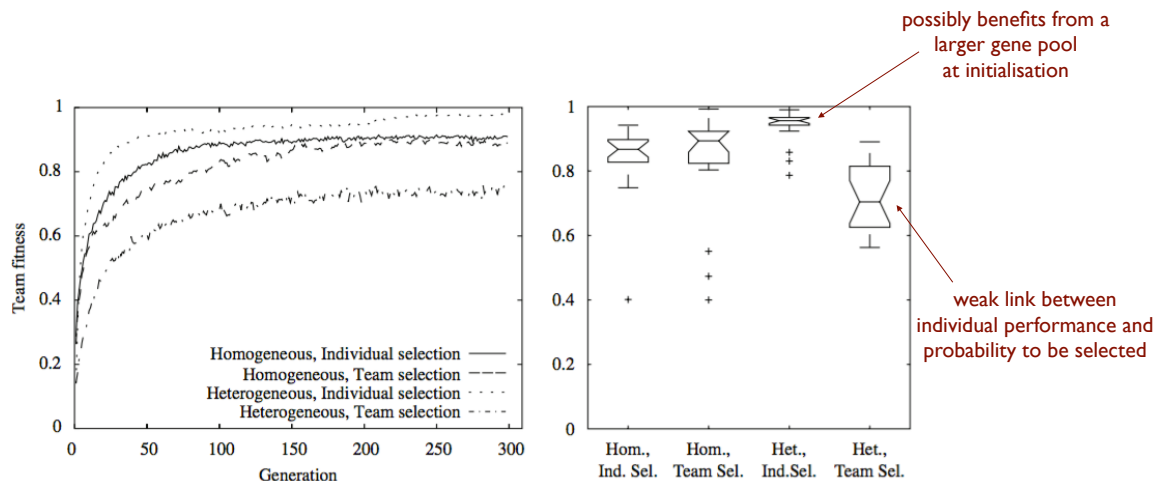
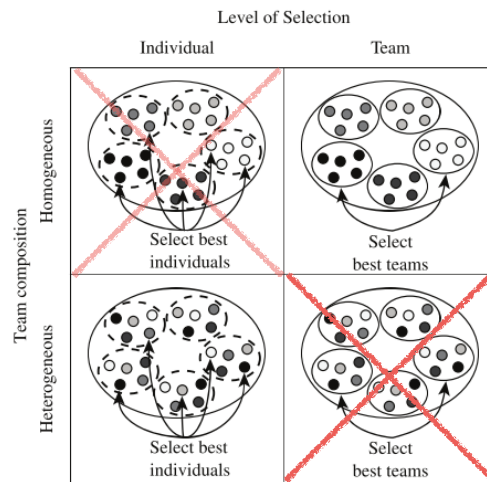


Fig. 5. Task 1—Individual Foraging. Left: Evolution of the best team fitness averaged over the best teams in 20 independent evolutionary runs over 300 generations. Right: The best team at generation 300 of each of the 20 independent experiments per evolutionary condition and per task was evaluated 1000 times. The mid-line in the box is the median, while the box represents the upper and lower quartile above and below the median. The bars outside the box generally represent the maximum and minimum values, except when there are outliers, which are shown as crosses. We define outliers as data points which differ more than 1.5 times the interquartile range from the border of the box. The notches represent the uncertainty in the difference of the medians for box-to-box comparison. Boxes whose notches do not overlap indicate that the medians differ at the 5% significance level [84]. In this task, which did not require cooperation, heterogeneous teams evolved with individual selection performed best, followed by homogeneous teams evolved with individual selection and homogeneous teams evolved with team selection. Heterogeneous teams evolved with team selection performed significantly worse than all other evolutionary conditions.

Control experiment: individual foraging

arena contains 6 small tokens ; each token brings 1 fitness point to forager

[Waibel et al., 2009]



- ✓ • Homogeneous teams w/ team selection is a safe choice
- ✗ • Homogeneous teams w/ indiv. selection is redundant (prefer the above)
- ✗ • Heterogeneous teams w/ team selection is inefficient
- ✓ • Heterogeneous teams w/ individual selection...
 - can benefit from a larger gene pool (fine if no cooperation needed)
 - may suffer from the credit assignment problem (may hinder coop.)

[Waibel et al., 2009]

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Take-home message on levels of selection

Team composition and level of selection matter

Homogeneous team is usually a safe choice

Heterogeneous team w/ ind. selection suffer from the credit assign. pb.

Q: Quels sont les combinaisons pertinentes?

1. sélectionner l'équipe lorsque le score est donnée au niveau de l'équipe
2. sélectionner l'individu lorsque le score est donnée au niveau de l'équipe
3. sélectionner l'équipe lorsque le score est donnée au niveau de l'individu
4. sélectionner l'individu lorsque le score est donnée au niveau de l'individu

Q: dans le cas de la sélection au niveau de l'individu et d'un score donné individuellement...

1. l'individu le mieux noté est effectivement plus performant que les autres
2. la performance du groupe est maximale lorsque aucun individu ne peut s'améliorer

Biological modelling

ER as an agent-based modelling tool for learning/evolution

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- Relevance as a modelling and simulation method
 - vs. mathematical modelling
 - simulates mechanistic aspects
 - vs. *in vitro* studies (or *in vivo* observations)
 - simulates longer evolutionary timescale

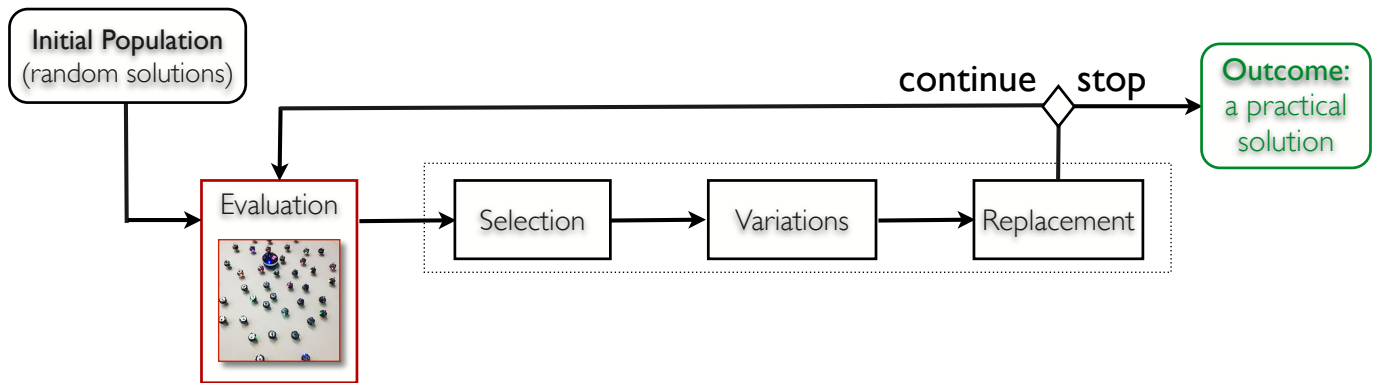
« So far, we have been able to study only one evolving system and we cannot wait for interstellar flight to provide us with a second. If we want to discover generalizations about evolving systems, we have to look at artificial ones. »

NATURE · VOL 355 · 27 FEBRUARY 1992

Byte-sized evolution

John Maynard Smith

Evolutionary robotics for modelling



- a model for evolution *and* social learning
 - innovation through variations, diffusion through selection
 - mechanistic and ecological aspects (e.g. physical interactions, resource scarcity)
- can be used to *force* population mixing and selection

[Mitri et al. 2012] [Doncieux et al., 2015]

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What are the benefits of **altruistic** behaviour ?

PLoS Biology | www.plosbiology.org

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May 2011 | Volume 9 | Issue 5 | e1000615

A Quantitative Test of Hamilton's Rule for the Evolution of Altruism

Markus Waibel^{1*}, Dario Floreano¹, Laurent Keller^{2*}

¹ Laboratory of Intelligent Systems, School of Engineering, Ecole Polytechnique Fédérale de Lausanne, Lausanne, Switzerland, ² Department of Ecology and Evolution, Biophore, University of Lausanne, Lausanne, Switzerland

Abstract

The evolution of altruism is a fundamental and enduring puzzle in biology. In a seminal paper Hamilton showed that altruism can be selected for when $rb - c > 0$, where c is the fitness cost to the altruist, b is the fitness benefit to the beneficiary, and r is their genetic relatedness. While many studies have provided qualitative support for Hamilton's rule, quantitative tests have not yet been possible due to the difficulty of quantifying the costs and benefits of helping acts. Here we use a simulated system of foraging robots to experimentally manipulate the costs and benefits of helping and determine the conditions under which altruism evolves. By conducting experimental evolution over hundreds of generations of selection in populations with different c/b ratios, we show that Hamilton's rule always accurately predicts the minimum relatedness necessary for altruism to evolve. This high accuracy is remarkable given the presence of pleiotropic and epistatic effects as well as mutations with strong effects on behavior and fitness (effects not directly taken into account in Hamilton's original 1964 rule). In addition to providing the first quantitative test of Hamilton's rule in a system with a complex mapping between genotype and phenotype, these experiments demonstrate the wide applicability of kin selection theory.

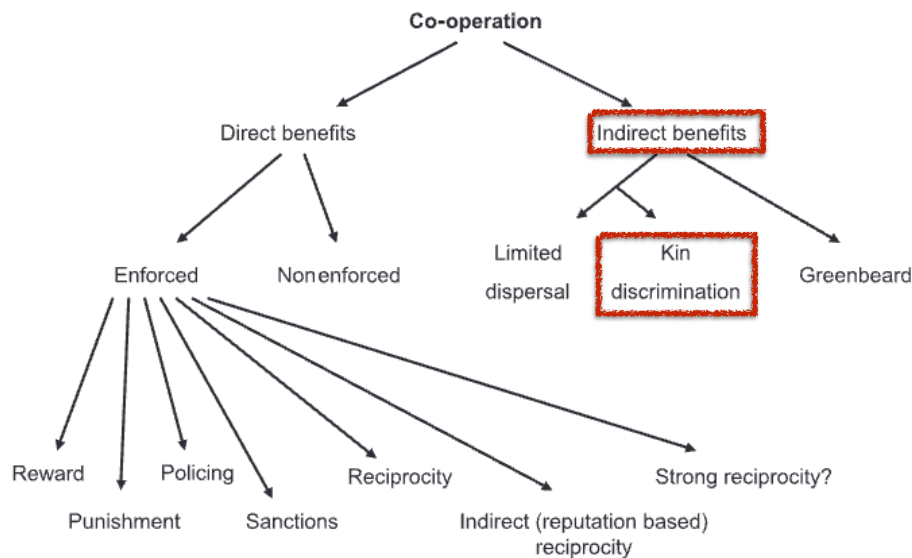


Fig. 1 A classification of the explanations for cooperation. Direct benefits explain mutually beneficial cooperation, whereas indirect benefits explain altruistic cooperation. Within these two fundamental categories, the different mechanisms can be classified in various ways – here we follow West *et al.* (2006; see also Sachs *et al.*, 2004; Lehmann & Keller, 2006). These possibilities are not mutually exclusive, for example a single act of cooperation could have both direct and indirect fitness benefits. We have listed some of the many different terms that have been used to describe the mechanisms for enforcing cooperation to emphasize that reciprocity (reciprocal altruism) is only one of many ways to obtain direct fitness benefits through cooperation. These enforcement mechanisms can also alter the indirect benefits of a behaviour (Lehmann & Keller, 2006), and determining the relationships between these terms remains an important task. Kin selection has been used to refer to (i) just those indirect benefits involving coancestry (i.e. limited dispersal and kin discrimination), or (ii) all indirect benefits (i.e. also including greenbeard effects).

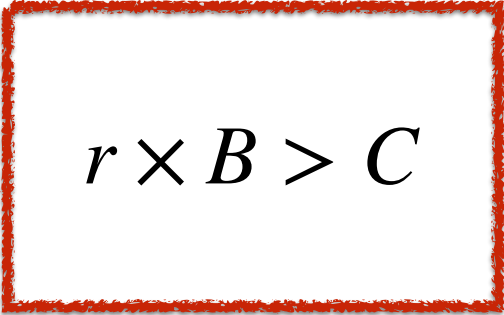
“if genes are identical by descent – by helping a close relative reproduce, an individual is still passing on its own genes to the next generation”

Quote from West et al. 2006, about the “narrow” definition of kin selection as originally given by Hamilton (1964)

“if genes are identical by descent – by helping a close relative reproduce, an individual is still passing on its own genes to the next generation”

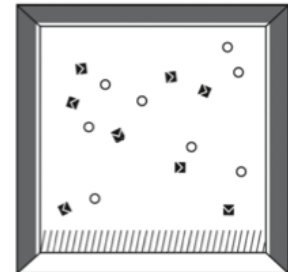
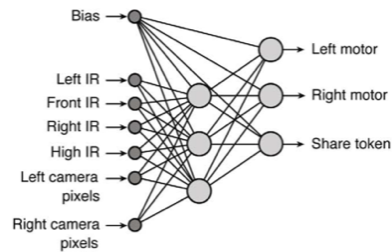
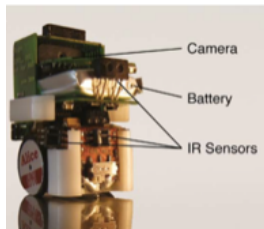
Quote from West et al. 2006, about the “narrow” definition of kin selection as originally given by Hamilton (1964)

Hamilton's rule (1964)


$$r \times B > C$$

r : genetic relatedness
B : reproductive benefit
C : reproductive cost

Does kin selection evolves?



8 robots, 8 items
objective: bring items to the wall
each robot decides to keep or share the reward

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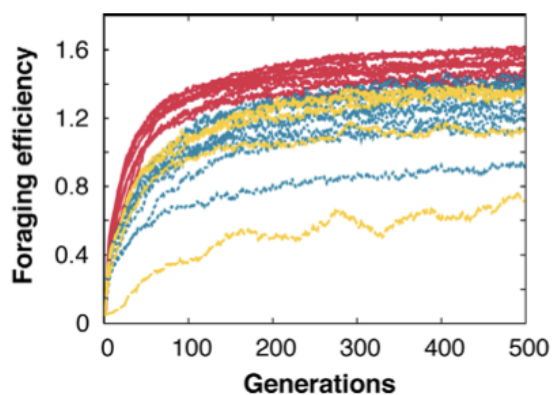
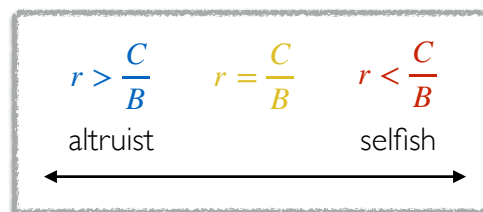


Figure 2. Mean group foraging efficiency during the 500 generations of selection. Lines represent the mean number of items transported successfully for each of the 25 treatments (five relatedness values $\times$ five c/b ratios, 20 replicates per treatment), with yellow dashed lines representing treatments where $r = c/b$, red lines treatments where $r < c/b$, and blue dotted lines treatments where $r > c/b$.
doi:10.1371/journal.pbio.1000615.g002

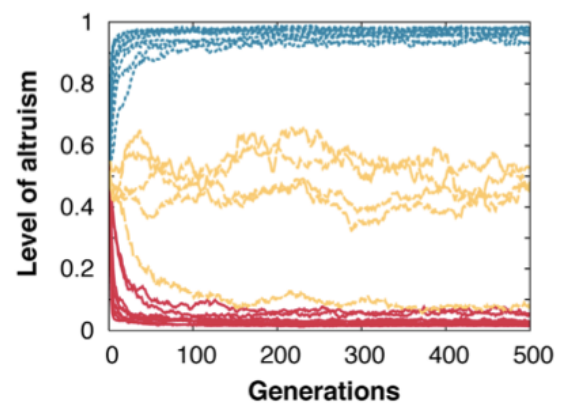


Figure 3. Mean level of altruism during the 500 generations of selection. Lines represent the mean level of altruism, defined as the proportion of food items shared with other group members, for each of the 25 treatments (five relatedness values $\times$ five c/b ratios, 20 replicates per treatment), with yellow dashed lines representing treatments where $r = c/b$, red lines treatments where $r < c/b$, and blue dotted lines treatments where $r > c/b$.
doi:10.1371/journal.pbio.1000615.g003

Take-home message

Evolutionary robotics can simulate a pseudo-realistic ecological process
in this case: effect on *inclusive* fitness

How difficult is it to **coordinate** to cooperate ?



To Cooperate or Not to Cooperate: Why Behavioural Mechanisms Matter

Arthur Bernard^{1*}, Jean-Baptiste André^{2☉}, Nicolas Bredeche^{1☉}

PLOS Computational Biology | DOI:10.1371/journal.pcbi.1004886 May 5, 2016

Abstract

Mutualistic cooperation often requires multiple individuals to behave in a coordinated fashion. Hence, while the evolutionary stability of mutualistic cooperation poses no particular theoretical difficulty, its evolutionary emergence faces a chicken and egg problem: an individual cannot benefit from cooperating unless other individuals already do so. Here, we use evolutionary robotic simulations to study the consequences of this problem for the evolution of cooperation. In contrast with standard game-theoretic results, we find that the transition from solitary to cooperative strategies is very unlikely, whether interacting individuals are genetically related (cooperation evolves in 20% of all simulations) or unrelated (only 3% of all simulations). We also observe that successful cooperation between individuals requires the evolution of a specific and rather complex behaviour. This behavioural complexity creates a large fitness valley between solitary and cooperative strategies, making the evolutionary transition difficult. These results reveal the need for research on biological mechanisms which may facilitate this transition.

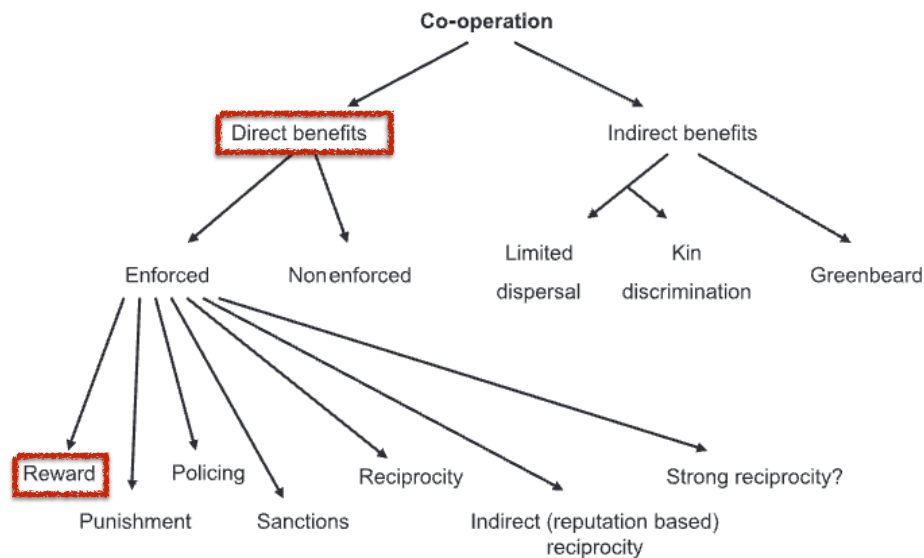


Fig. 1 A classification of the explanations for cooperation. Direct benefits explain mutually beneficial cooperation, whereas indirect benefits explain altruistic cooperation. Within these two fundamental categories, the different mechanisms can be classified in various ways – here we follow West *et al.* (2006; see also Sachs *et al.*, 2004; Lehmann & Keller, 2006). These possibilities are not mutually exclusive, for example a single act of cooperation could have both direct and indirect fitness benefits. We have listed some of the many different terms that have been used to describe the mechanisms for enforcing cooperation to emphasize that reciprocity (reciprocal altruism) is only one of many ways to obtain direct fitness benefits through cooperation. These enforcement mechanisms can also alter the indirect benefits of a behaviour (Lehmann & Keller, 2006), and determining the relationships between these terms remains an important task. Kin selection has been used to refer to (i) just those indirect benefits involving coancestry (i.e. limited dispersal and kin discrimination), or (ii) all indirect benefits (i.e. also including greenbeard effects).

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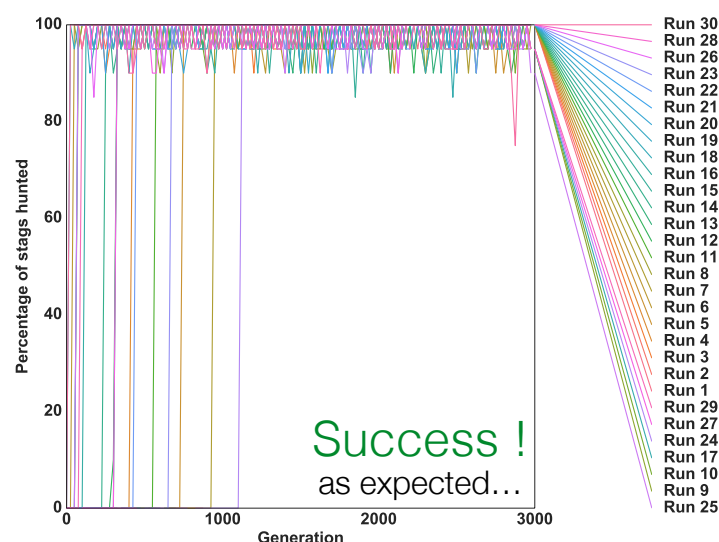
from: West et al. 2006 “social semantics”

The Stag Hunt Problem

[Skyrms, 2004]

You hunt \ Partner hunts		
	Hare	Stag
Hare	1 ; 1	1 ; 0
Stag	0 ; 1	2 ; 2

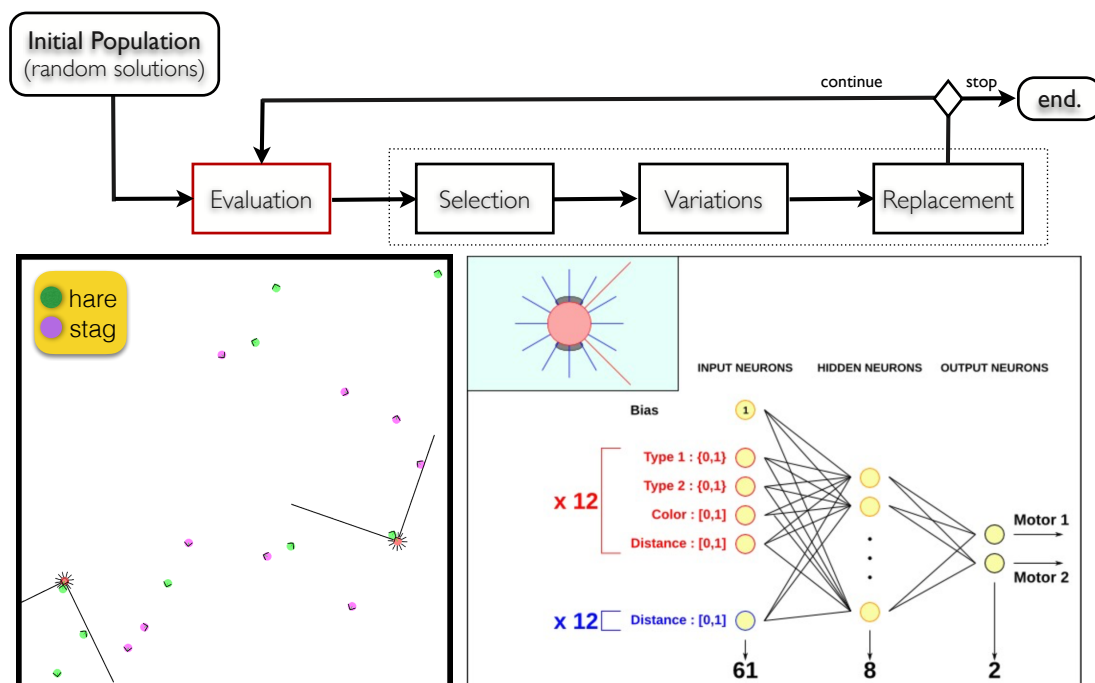
Risk-dominant equilibrium
 Payoff-dominant equilibrium



Numerical simulation with an initial population of hare hunters (i.e. worst case)

Open question

cooperation is beneficial and looks easy...
...why are there so few examples in Nature?

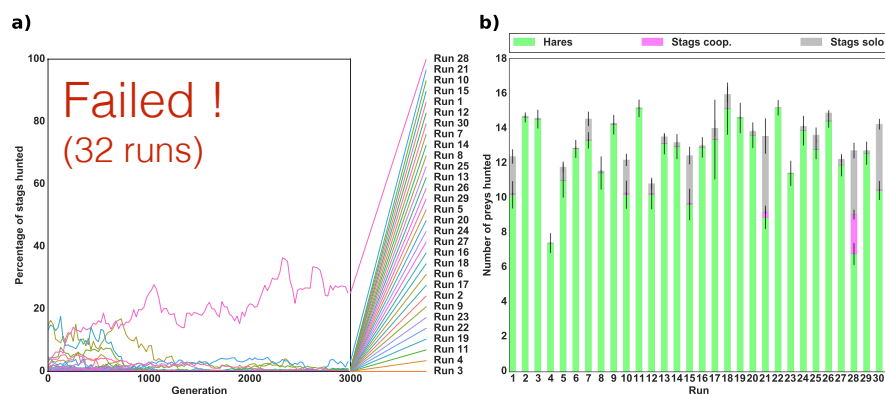
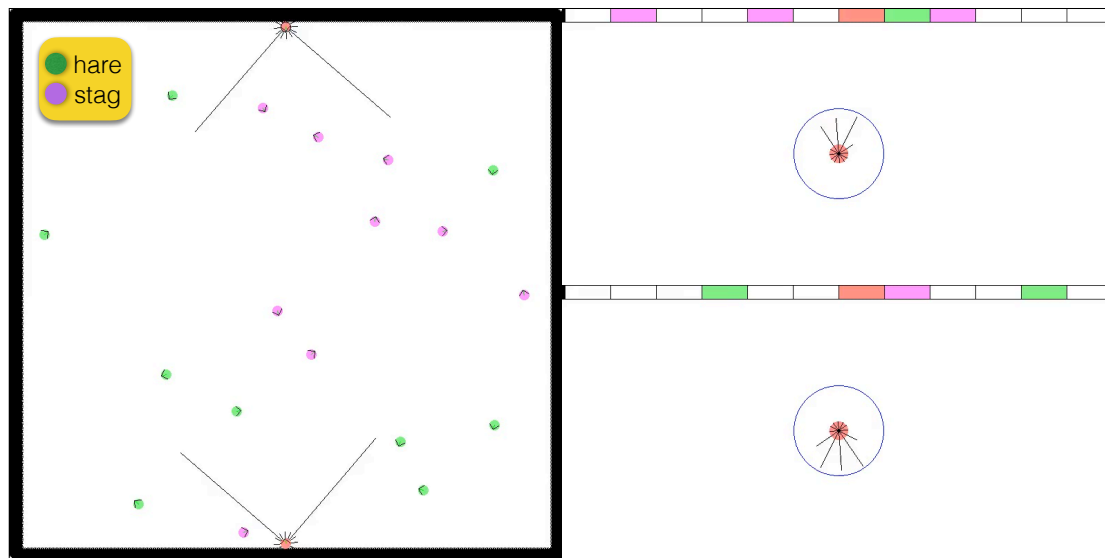


Evolution :

- population of 100 agents
- tournament selection

Evaluation :

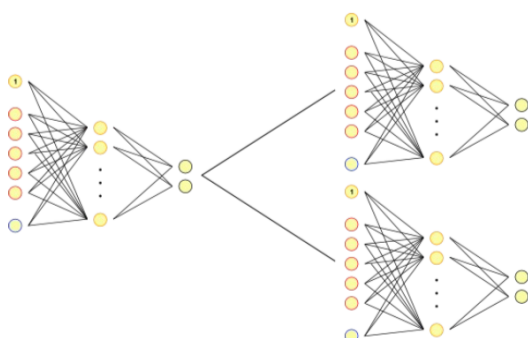
- 2 types of static resource (9 "hare" + 9 "stags") ; Payoffs: Hare << Stag
- robots are *paired* (i.e. only two robots in the arena)
- performance of an individual is the average of 5 random pairing (i.e. limited mixing)



Breaking behavioural symmetry

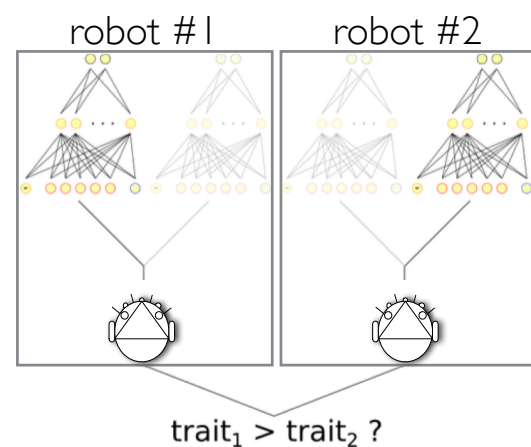
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New evolutionary operator:
network duplication

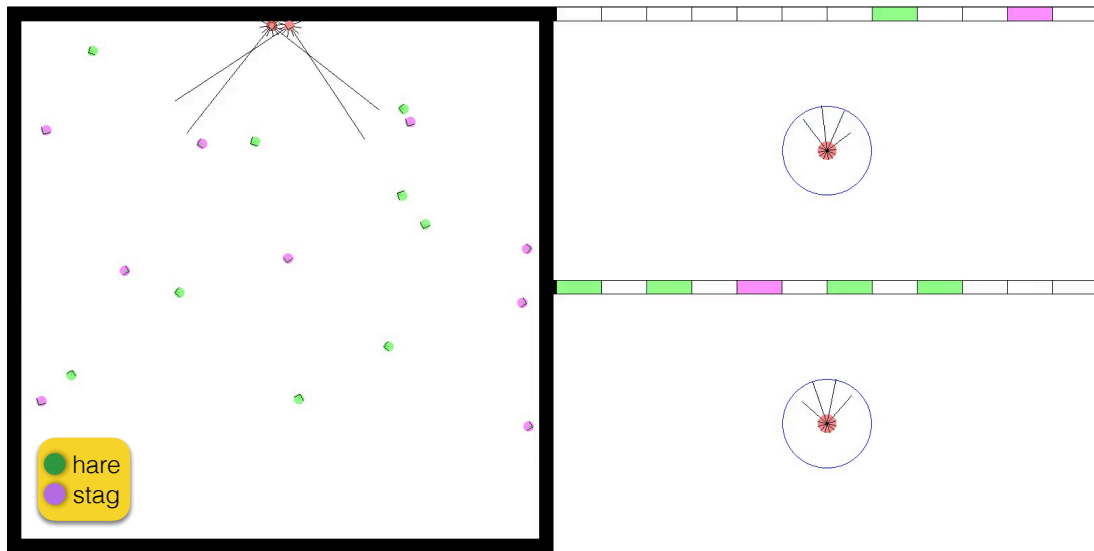


A network can be duplicated
A network can be deleted
Either 1 or 2 network(s)
 $p_{\text{duplicate}} = 5 \times 10^{-2}$
 $p_{\text{delete}} = 5 \times 10^{-3}$

During evaluation:
each robot chooses a network



Two strategies tested:
(1) networks are chosen for the evaluation
(2) networks are chosen at each prey capture



“Leader-Follower”
coordination **with** specialised roles

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Partner hunts You hunt		Leader hunts...	
		Hare	Stag
Hare	1 ; 1	Leader/follower	1 ; 1
Stag	2 ; 2		2 ; 2

coordination game

single player optimisation problem

- Mechanistic constraints change the rule of the game
 - The transition to a payoff-dominant strategy can be obtained as the by-product of individual optimisation
 - A certain level of “cognitive” complexity may be required for symmetry breaking

Individual-level adaptation leads to **collective behaviours**

Take-home message

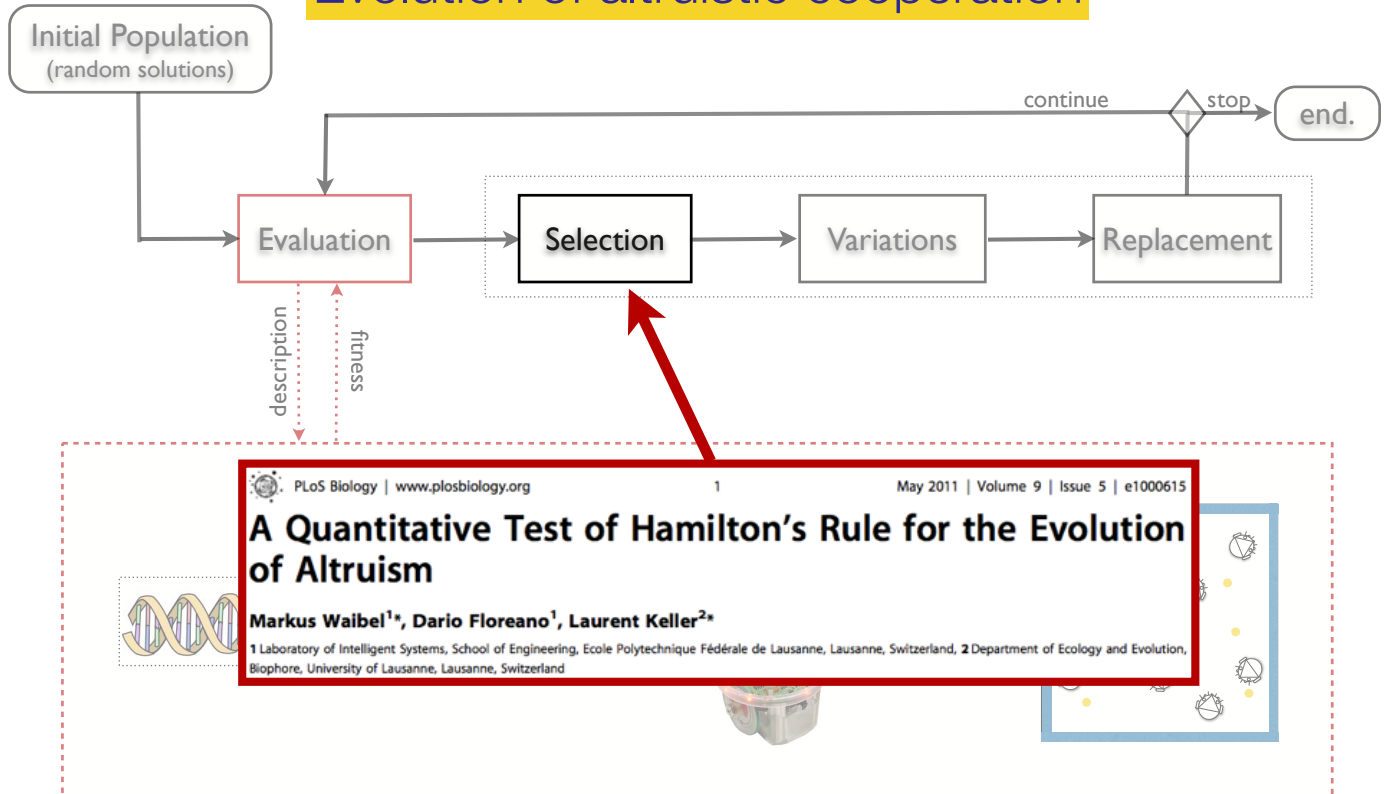
Mechanistic constraints can play a role in evolutionary dynamics

e.g.: critical to understand that individual-level adaptation leads to collective behaviours

Conclusion on collective ER as a modelling method

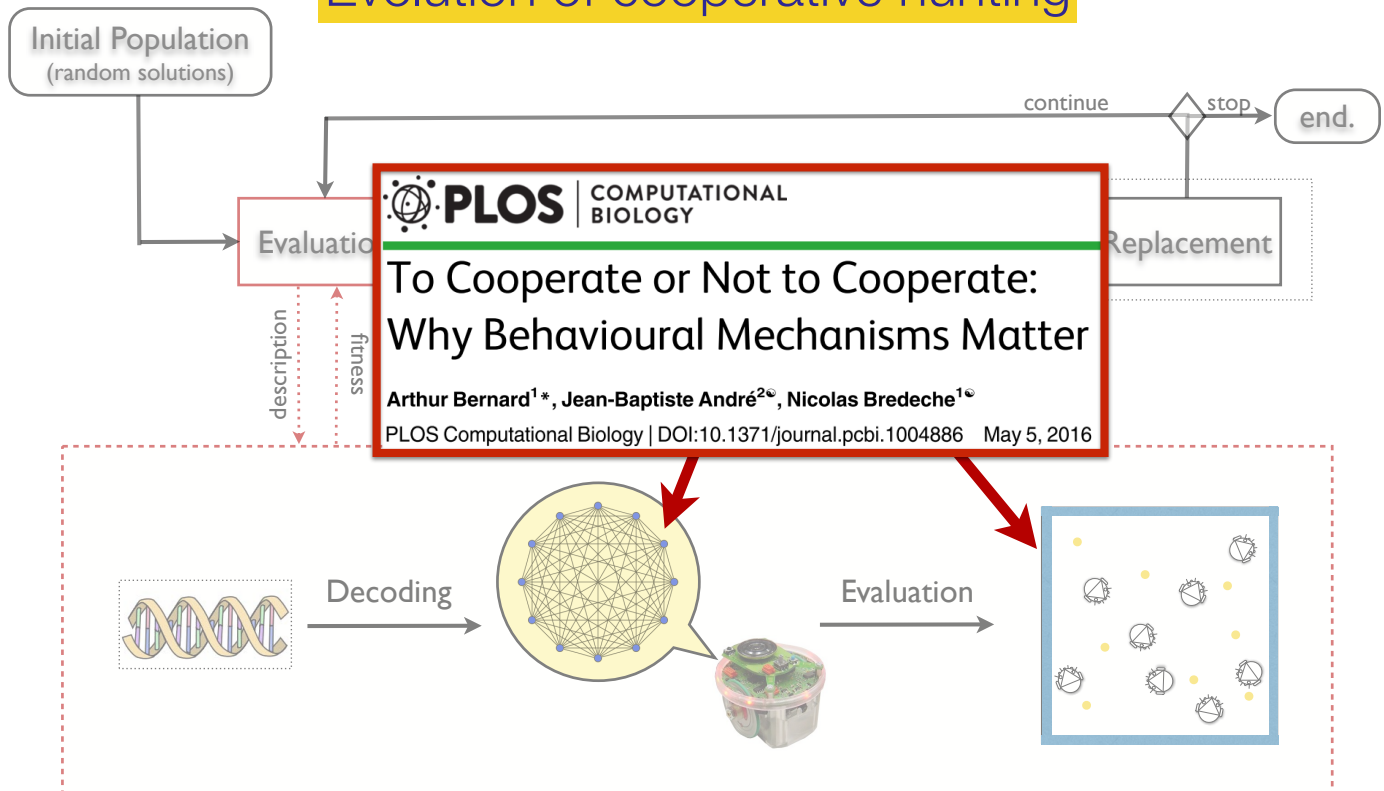
- Wrapping up
 - ▶ relevant to try out new hypothesis (validation in the wild is another matter)
 - ▶ lessons learned also benefits ER as an engineering method
 - ▶ all aspects of the algorithm can be controlled

Evolution of altruistic cooperation



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Evolution of cooperative hunting



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Evolvability

Initial Population
(random solutions)

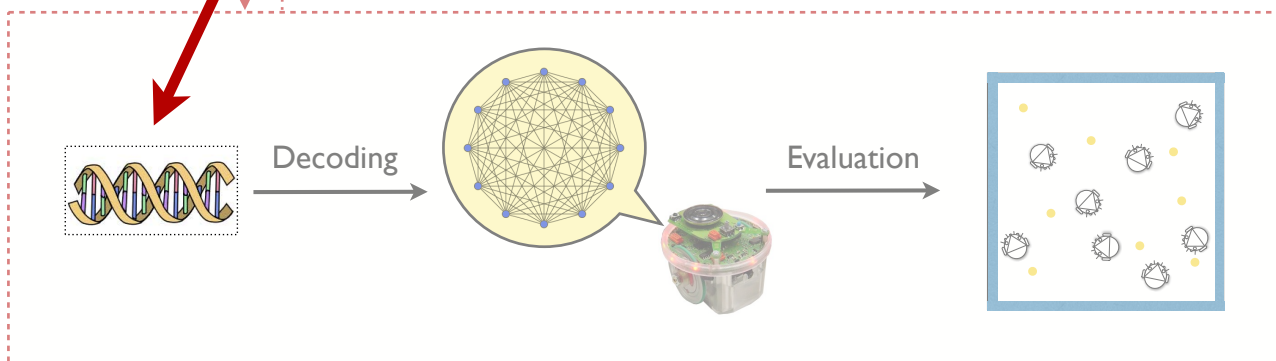
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Evolvability Is Inevitable: Increasing Evolvability without the Pressure to Adapt

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Evolution of signalling

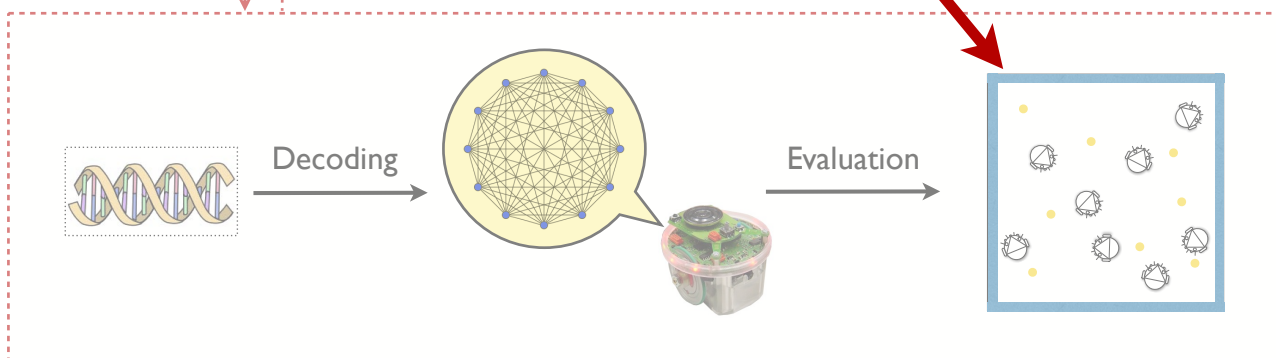
Initial Population
(random solutions)

Historical contingency affects signaling strategies and competitive abilities in evolving populations of simulated robots

Steffen Wischmann^{a,b}, Dario Floreano^b, and Laurent Keller^{a,1}

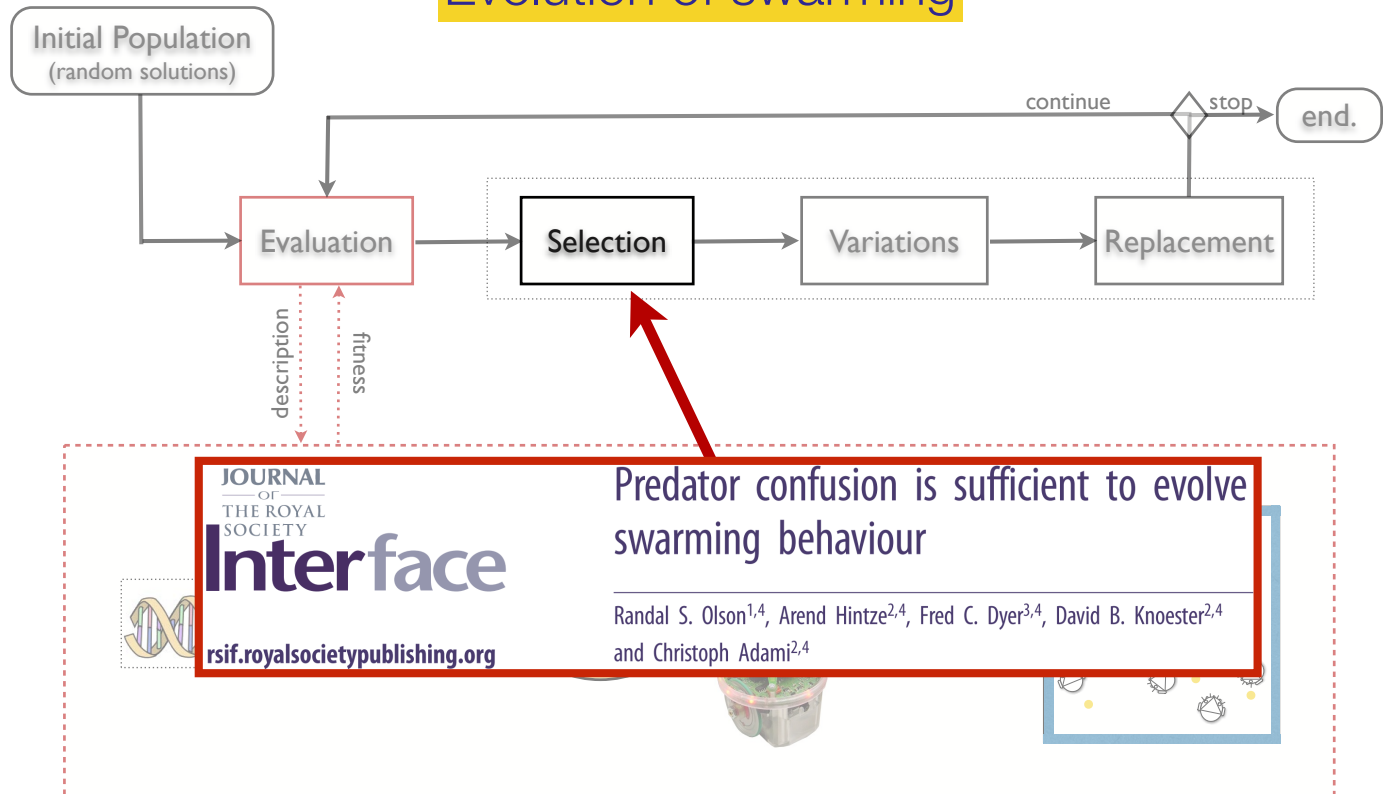
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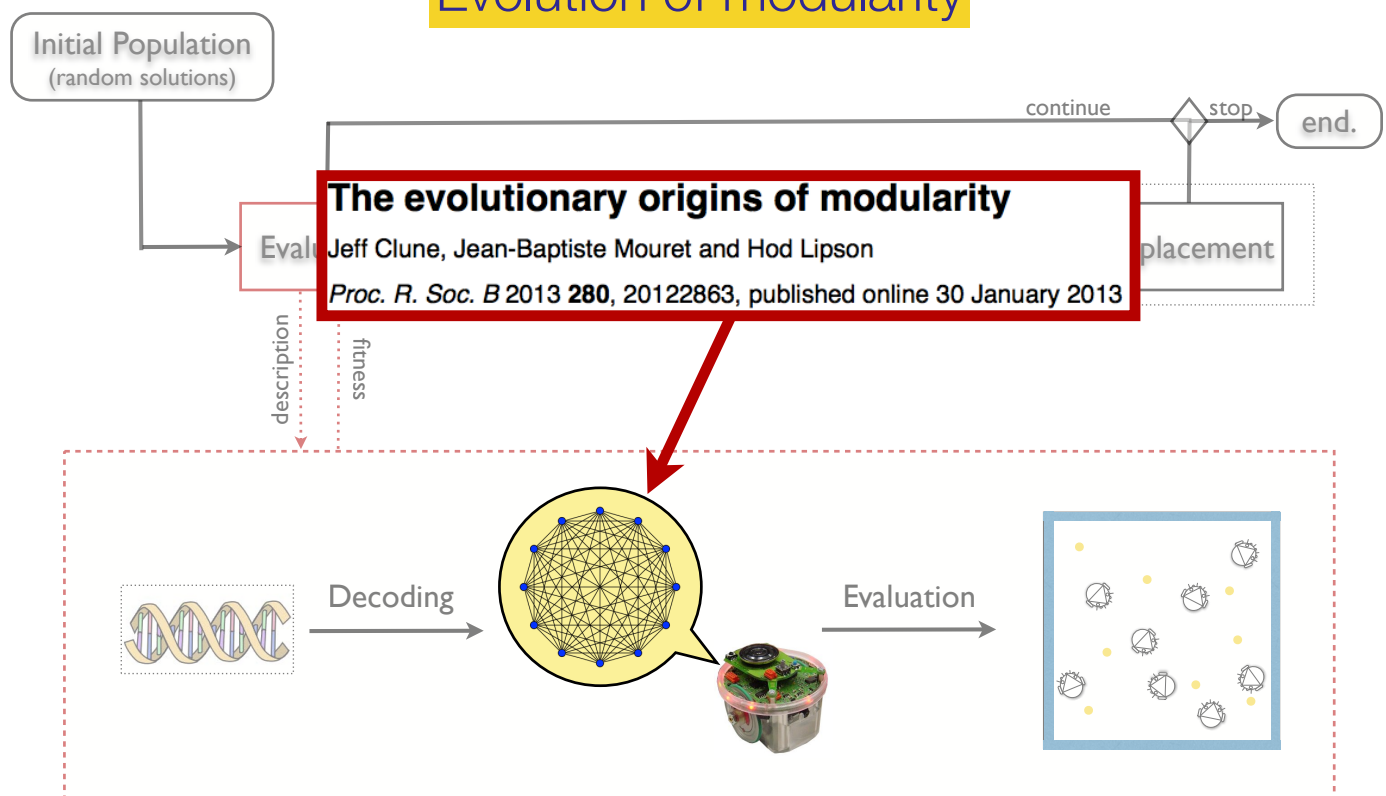
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Evolution of swarming



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Evolution of modularity



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Q: Quel intérêt d'utiliser la robotique collective évolutionnaire pour la biologie?

1. cela permet de prendre en compte des contraintes plus réalistes
2. cela permet d'expliquer des dynamiques observées dans la nature
3. cela permet de modéliser des architectures cognitives complexes
4. cela permet d'affiner des hypothèses sur les dynamiques adaptatives

Q: La chasse en groupe peut être expliquée par...

1. de la coopération mutualiste
2. de l'altruisme

Conclusions

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- Evolutionary robotics for collectives
 - Evolutionary dynamics in competitive setups
 - Large collectives
 - ▶ levels of selection
 - ▶ w.r.t. specialisation
 - clones can display specialised behaviours
 - non-clonal approach intrinsically favours specialisation
 - A model to study evolutionary/learning dynamics